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Hybrid DAG-Edge-IoT Temperature-Credit Token Model for Routing Optimization of Micro Cold-Storage in the TNBBS Conservation Corridor

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ABSTRACT
Background: Transporting temperature-sensitive perishable catch across remote roads in West Lampung and TNBBS incurs high thermal degradation risks and fragmented logistics accounting. Objectives: To develop a Directed Acyclic Graph (DAG) ledger coupled with Edge-IoT telemetry and tokenized temperature credits for decentralized routing. Methods: Low-power Edge-IoT telemetry nodes measuring compressor duty cycles and GPS routing across 11 sub-districts of Pesisir Barat. Results: DAG transaction throughput achieves near-zero confirmation latency with cryptographic verification of thermal integrity, incentivizing local couriers through tokenized fuel-saving credits. Conclusion: The architecture bridges the technological divide in 3T archipelagic corridors through decentralized, tamper-proof cold-chain verification. Published in: Journal of Coastal Frontiers & Applied Development (JCFAD) , Vol. 1 No. 1 (2026), pp. 1–12. Official publisher: PT CHAIN ASIA CONECT Research & Development Division.

Keywords: Pesisir Barat, Krui, Frontier Economics, Legal Certainty, Indonesian Civil Code, Empirical Modeling, Cold Storage.

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1. Introduction

1.1 Empirical Background

The South Barisan Mountain National Park (TNBBS) region is a conservation zone characterized by high biodiversity and a sensitive microclimate. Logistics activities, particularly the transportation of agricultural produce and pharmaceuticals, demand a micro-cold-storage system capable of maintaining a stable temperature below 5 °C. However, the cooling infrastructure in this area still relies on centralized systems that incur high energy costs and pose a risk of temperature failure, which can damage products and threaten the ecosystem.

1.2 Literature Review

Several prior studies have examined the use of blockchain for supply-chain traceability (Kumar & Lee, 2021) and the application of reinforcement-learning algorithms for vehicle routing (Zhang et al., 2022). Static thermal models for cold-storage remain dominant (Mendoza et al., 2020). Meanwhile, Directed Acyclic Graph (DAG) technology has been shown to reduce energy consumption compared with proof-of-work (Böhme & Gervais, 2021). Nevertheless, the integration of cooling-capacity tokenisation (temperature-credit) with legal mechanisms has not yet been explored.

1.3 State of the Art & Research Gap

The current state of the art includes: (a) proof-of-work (PoW)-based blockchain for smart contracts, (b) single-agent artificial intelligence for route optimisation, and (c) token-economy models that do not link token value to physical temperature parameters. The research gap lies in the lack of a model that simultaneously (i) employs DAG for low-energy consensus, (ii) implements temperature-credit tokens whose value is influenced by actual temperature, (iii) utilises multi-agent reinforcement learning (MARL) for dynamic routing, and (iv) incorporates a legal framework that binds temperature penalties to token value within a conservation context.

1.4 Novelty of the Research

This study introduces the Hybrid DAG-Edge-IoT Temperature-Credit Token (TCT) Model, which combines four innovative elements: (1) a DAG-based ledger for low-cost transactions, (2) edge-IoT sensors that measure temperature in real time, (3) tokenisation of cooling capacity (TCT) with smart contracts that adjust token value based on temperature deviation, and (4) a meta-legal contract that links temperature penalties to indemnity clauses in the Indonesian Civil Code (KUHPerdata). This model generates a decentralised, adaptive, sustainable, and regulation-compliant cooling market.

1.5 Problem Statement

1. How can a hybrid DAG-Edge-IoT architecture be designed to accommodate real-time temperature-credit transactions?
2. How can a MARL algorithm be developed for micro-cold-storage routing that minimises temperature deviation and energy cost?
3. How can a meta-legal contract be drafted that links temperature penalties to token value in accordance with KUHPerdata?
4. To what extent can this model be implemented technically, economically, and regulatory-wise within the TNBBS conservation corridor?

2. Research Methodology

2.1 Juridical-Empirical / Socio-Legal Approach

4. Conclusion and Policy Implications

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References

- Bappenas. (2023). **Strategic Plan for 3T Regional Development 2020-2024**. Ministry of Finance/PPN/Bappenas RI.
- BPS West Coast Regency. (2023). **West Coast Regency in Figures 2023: Tourism & Transportation Statistics**.
- Christopher, M., & Peck, H. (2024). Building resilient supply networks in remote clusters. **International Journal of Logistics Management**, 35(1), 1-24.
- Govindan, K., & Soleimani, H. (2022). Reverse logistics and closed-loop micro-storage in coastal areas. **Journal of Cleaner Production**, 98, 3-11.
- Gunawan, J., & Siregar, F. (2022). Reconstruction of legal protection for SMEs in supply contracts. **Journal of Business Law Research**, 9(2), 215-228.
- Hartono, S., & Wahyudi, T. (2023). Legal certainty of force majeure clauses in disaster-prone areas. **Journal of Law & Governance**, 6(1), 45-58.
- Subekti, R. (2020). **Contract Law and the Principle of Freedom of Contract**. Jakarta: Intermasa.
- Widodo, K. H., & Purwaditya, A. (2021). Sustainable logistics for remote islands connectivity. **Indonesian Journal of Geography**, 53(2), 260-271.

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RESEARCH INTEGRITY & OPEN ACCESS DECLARATION

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REFERENCES

- Bappenas. (2023). *Rencana Strategis Pembangunan Daerah 3T 2020-2024*. Kementerian PPN/Bappenas RI.
- BPS Kabupaten Pesisir Barat. (2023). *Kabupaten Pesisir Barat Dalam Angka 2023: Statistik Pariwisata & Transportasi*.
- Christopher, M., & Peck, H. (2024). Building resilient supply networks in remote clusters. *Int. J. Logistics Mgmt*, 35(1), 1-24.
- Govindan, K., & Soleimani, H. (2022). Reverse logistics and closed-loop micro-storage in coastal areas. *J. Cleaner Prod.*, 98, 3-11.
- Gunawan, J., & Siregar, F. (2022). Rekonstruksi Perlindungan Hukum Bagi UMKM dalam Kontrak Pasokan. *Jurnal Riset Hukum Bisnis*, 9(2), 215-228.
- Hartono, S., & Wahyudi, T. (2023). Kepastian Hukum Klausul Force Majeure di Kawasan Rawan Bencana. *J. Hukum & Tata Kelola*, 6(1), 45-58.
- Subekti, R. (2020). *Hukum Perjanjian dan Asas Kebebasan Berkontrak*. Jakarta: Intermasa.
- Widodo, K. H., & Purwaditya, A. (2021). Sustainable Logistics for Remote Islands Connectivity. *Indonesian J. Geography*, 53(2), 260-271.